

Saduyev Daniyal

LAB_WORK №1

EmpID	SSN	Email	Phone	Name	Department	Salary
101	123-45-6789	john@company.com	555-0101	John	IT	75000
102	987-65-4321	mary@company.com	555-0102	Mary	HR	68000
103	456-78-9123	bob@company.com	555-0103	Bob	IT	72000

TASK №1.

Superkey are:

{EmpID}, {EmpID,SSN}, {SSN}, {Email}, {EmpID,SSN,Email,Phone}, {Email,phone}.

All candidate keys:

{EmpID}, {SSN}, {Email}.

Primary key and why:

{EmpID}, I would choose EmpID as the primary key because it is a surrogate identifier. Unlike SSN, which is country-specific, EmpID can be used universally. It also does not change over time, unlike an email address, and most importantly, it is unique for every employee.

4. Can two employees have the same phone number? Justify your answer based on the data shown.

Based on the data shown, all employees have different phone numbers. However, the data does not prove that phone numbers must always be unique, so two employees could potentially have the same phone number. For example, the task does not specify that the phone number must be a personal mobile number. Two employees may live together and provide the same home phone number. Therefore, Phone cannot be guaranteed to be unique. In addition, a phone number is sensitive personal information and may change over time, so it is not a good choice for a primary key.

TASK №2.

[Relation B: Course Registration](#)

Registration(StudentID, CourseCode, Section, Semester, Year, Grade, Credits)

TASK:

- 1) Determine the minimum attributes needed for the primary key:
{StudentID, CourseCode, Section, Semester, Year}
- 2) Explain why each attribute in your primary key is necessary :

Each attribute in the primary key is necessary because StudentID alone is not enough to identify one registration. A student can take different courses, study in different sections, and take courses in different semesters and years. Therefore, CourseCode, Section, Semester, and Year are also needed to uniquely identify a registration.

- 3) Identify any additional candidate keys (if they exist):

There are no additional candidate keys based on the given business rules. Grade is not unique because different students can receive the same grade. Credits are also not unique because different courses can have the same number of credits. The other individual attributes are not enough to uniquely identify one registration.

Task 1.2: Foreign Key Design

- 1) Student.advisorID refers to Professor.ProfID
- 2) Course.DepartmentCode refers to Department.DeptCode
- 3) Enrollment.StudentID refers to Student.studentID
- 4) Enrollment.CourseID refers to Course.CourseID

Part 2: ER Diagram Construction.

1. Identify all entities (specify which are strong and which are weak):
Strong: Patients, Doctors, Department
Weak: Hospital_room, Appointments, Prescriptions
2. Identify all attributes for each entity (classify as simple, composite, multi-valued, or derived):

Patient: PatientID — simple, key; Name — simple; Birthdate — simple; Address — composite (Street, City, State, Zip); PhoneNumber — multi-valued; InsuranceInformation — simple.

Doctor: DoctorID — simple, key; Name — simple; Specialization — multi-valued; PhoneNumber — simple; OfficeLocation — simple.

Department: DepartmentCode — simple, key; Name — simple; Location — simple.

Appointment: DateTime — simple; Purpose — simple; Notes — simple.

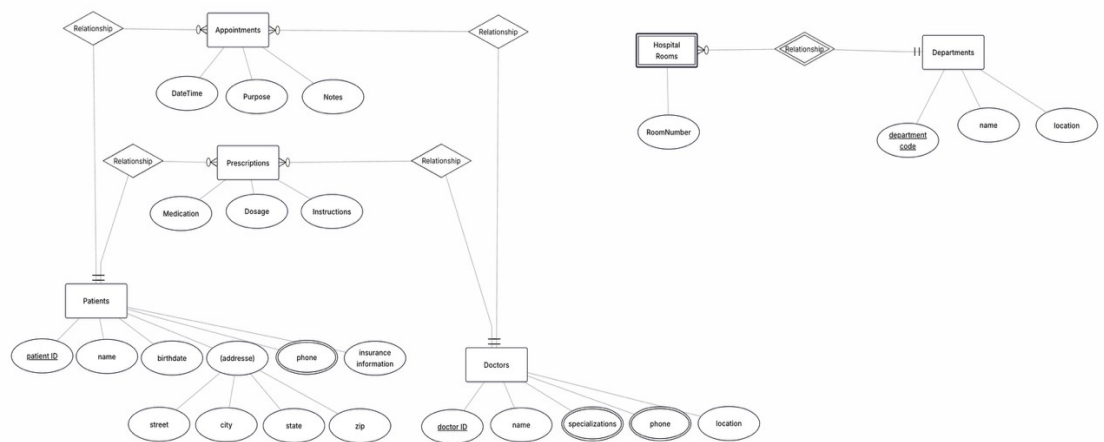
Prescription: Medication — simple; Dosage — simple; Instructions — simple.

HospitalRoom: RoomNumber — simple; together with DepartmentCode it identifies the room.

- Identify all relationships with their cardinalities (1:1, 1:N, M:N):

Relationships and cardinalities: Patient–Appointment — 1:N; Doctor–Appointment — 1:N; Patient–Prescription — 1:N; Doctor–Prescription — 1:N; Department–HospitalRoom — 1:N.

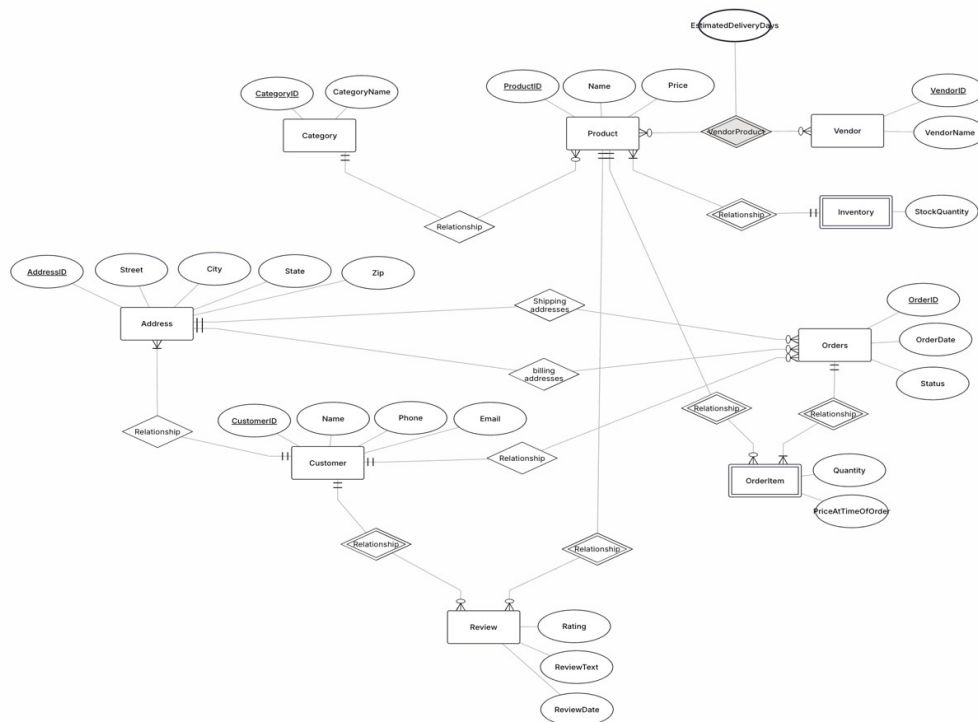
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- Mark primary keys:

Primary Keys: Patient — PatientID; Doctor — DoctorID; Department — DepartmentCode; Appointment — AppointmentID; Prescription — PrescriptionID; HospitalRoom — composite key (DepartmentCode, RoomNumber).

Task 2.2: E-commerce Platform



2) OrderItem must contain OrderID and ProductID as a composite primary key, because these values identify the specific product within a specific order. Quantity and PriceAtTimeOfOrder alone are not enough to uniquely identify one OrderItem.

3) Vendor and Product have a many-to-many relationship because one vendor can supply many products, and one product can be supplied by many vendors. This relationship needs the attribute EstimatedDeliveryDays to store the approximate delivery time for each vendor-product pair.

Part 4: Normalization Workshop

Task 4.1: Denormalized Table Analysis

StudentID → StudentName, StudentMajor

ProjectID → ProjectTitle, ProjectType, SupervisorID, StartDate, EndDate

SupervisorID → SupervisorName, SupervisorDept

(StudentID, ProjectID) → Role, HoursWorked

2)

Redundancy: Student, project, and supervisor data can be repeated in many rows.

Update anomaly: If a supervisor changes department, the same information must be updated in every related row.

Insert anomaly: A new project cannot be added properly if no student is assigned to it yet.

Delete anomaly: If the last student is removed from a project, information about that project and its supervisor may also be lost.

3) The table has no explicit 1NF violations because all attributes appear to contain atomic values. If an attribute contained multiple values in one cell, I would separate those values into individual rows or create a separate related table.

4) Apply 2NF

Primary Key:

(StudentID, ProjectID)

Partial Dependencies:

StudentID → StudentName, StudentMajor

ProjectID → ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, StartDate, EndDate

These are partial dependencies because they depend only on part of the composite primary key.

Role and HoursWorked depend on both StudentID and ProjectID.

2NF Decomposition:

Student(StudentID, StudentName, StudentMajor)

Project(ProjectID, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, StartDate, EndDate)

StudentProject(StudentID, ProjectID, Role, HoursWorked)

5) Apply 3NF

Transitive dependency:

$\text{SupervisorID} \rightarrow \text{SupervisorName}, \text{SupervisorDept}$

$\text{ProjectID} \rightarrow \text{SupervisorID}$

Therefore, SupervisorName and SupervisorDept depend on ProjectID through SupervisorID.

3NF Decomposition:

Student(StudentID, StudentName, StudentMajor)

Supervisor(SupervisorID, SupervisorName, SupervisorDept)

Project(ProjectID, ProjectTitle, ProjectType, SupervisorID, StartDate, EndDate)

StudentProject(StudentID, ProjectID, Role, HoursWorked)

This removes the transitive dependency and keeps each type of information in its own table.

Task 4.2: Advanced Normalization

1. Primary Key

The primary key is:

(StudentID, CourseID)

because a student can be enrolled in multiple courses, and each course can have multiple students.

2. Functional Dependencies

$\text{StudentID} \rightarrow \text{StudentMajor}$

$\text{CourseID} \rightarrow \text{CourseName}$

$\text{InstructorID} \rightarrow \text{InstructorName}$

$\text{CourseID} \rightarrow \text{InstructorID}, \text{TimeSlot}, \text{Room}$

$(\text{Room}, \text{TimeSlot}) \rightarrow \text{Building}$

3. BCNF Check

The table is not in BCNF because some determinants are not superkeys.

For example:

$\text{StudentID} \rightarrow \text{StudentMajor}$

StudentID is not a superkey of the whole table, so this violates BCNF.

The same problem exists with CourseID and InstructorID.

4. BCNF Decomposition

Student(StudentID, StudentMajor)

Course(CourseID, CourseName)

Instructor(InstructorID, InstructorName)

CourseSection(CourseID, InstructorID, TimeSlot, Room)

RoomSchedule(Room, TimeSlot, Building)

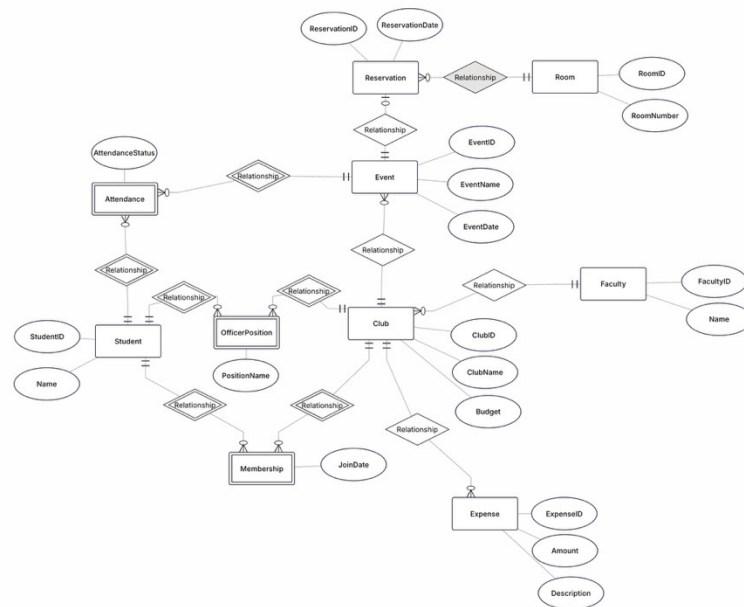
Enrollment(StudentID, CourseID)

5. Loss of Information

The decomposition should be lossless if the tables are joined using their keys and foreign keys. However, the original table does not contain a separate SectionID, so if one course can have multiple sections, some information about the exact section may be ambiguous. A separate SectionID would make the design clearer and safer.

[Part 5: Design Challenge](#)

Task 5.1: Real-World Application



2) Student(
 StudentID PK,
 Name,
 Email
)

Faculty(
 FacultyID PK,
 Name
)

Club(
 ClubID PK,
 ClubName,
 Budget,

AdvisorID FK
)

Membership(
StudentID PK, FK,
ClubID PK, FK,
JoinDate
)

Event(
EventID PK,
ClubID FK,
EventName,
EventDate
)

Attendance(
StudentID PK, FK,
EventID PK, FK
)

OfficerPosition(
StudentID PK, FK,
ClubID PK, FK,
PositionName
)

Room(

RoomID PK,
RoomNumber
)

Reservation(
ReservationID PK,
EventID FK,
RoomID FK
)

Expense(
ExpenseID PK,
ClubID FK,
Amount,
Description
)

3) **Design Decision:**

I used a separate Membership table for the relationship between Student and Club because it allows us to store extra information such as the date when the student joined the club.

4 task.

1. Find all students who are members of the Robotics Club.
2. List all club events scheduled for next week with their room reservations.
3. Show all expenses of the Computer Science Club for the current semester.